



Java Data Types

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Java Data Types

As explained in the previous chapter, a variable in Java must be a specified data type:

Example

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```
int myNum = 5;           // Integer (whole number)
float myFloatNum = 5.99f; // Floating point number
char myLetter = 'D';     // Character
boolean myBool = true;   // Boolean
String myText = "Hello"; // String
```

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Data types are divided into two groups:

- Primitive data types - includes `byte`, `short`, `int`, `long`, `float`, `double`, `boolean` and `char`
- Non-primitive data types - such as `String`, `Arrays` and `Classes` (you will learn more about these in a later chapter)

Primitive Data Types

A primitive data type specifies the type of a variable and the kind of values it can hold.

There are eight primitive data types in Java:



short	Stores whole numbers from -32,768 to 32,767
int	Stores whole numbers from -2,147,483,648 to 2,147,483,647
long	Stores whole numbers from -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
float	Stores fractional numbers. Sufficient for storing 6 to 7 decimal digits
double	Stores fractional numbers. Sufficient for storing 15 to 16 decimal digits
boolean	Stores true or false values
char	Stores a single character/letter or ASCII values

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You Cannot Change the Type

Once a variable is declared with a type, it cannot change to another type later in the program:

Example

```
int myNum = 5;           // myNum is an int
// myNum = "Hello";     // Error: cannot assign a String to an int

String myText = "Hi";    // myText is a String
// myText = 123;         // Error: cannot assign a number to a String
```

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If you really need to change between types, you must use type casting or conversion methods (for example, turning an **int** into a **double**).

Exercise [?]

Drag and drop the correct integer value to complete the code.

```
int myNum =    10    ;  
System.out.println(myNum);
```

"10"

10.5

true

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Video: Java Data Types